



Important: Text generated by AI and may contain incorrect information, always check the content.

```
```python
Import necessary libraries
import matplotlib.pyplot as plt
import seaborn as sns

Check the presence of null values
display(df.isnull().sum())

Check the distribution of categorical variables
categorical_cols = ['GENDER', 'LUNG_CANCER']
for col in categorical_cols:
 display(df[col].value_counts())

Plot the distribution of categorical variables
plt.figure(figsize=(10, 4))
for i, col in enumerate(categorical_cols):
 plt.subplot(1, 2, i + 1)
 sns.countplot(x=df[col])
 plt.title(f'Distribution of {col}')
plt.tight_layout()
plt.show()

Check the distribution of numerical variables
numerical_cols = df.select_dtypes(include=['int64']).columns.tolist()
display(df[numerical_cols].describe())

Plot the distribution of numerical variables
plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_cols):
 plt.subplot(4, 4, i + 1)
 sns.histplot(df[col], kde=True)
 plt.title(f'Distribution of {col}')
plt.tight_layout()
plt.show()

Plot the correlation matrix
plt.figure(figsize=(12, 10))
sns.heatmap(df[numerical_cols].corr(), annot=True, cmap='coolwarm', square=True)
plt.title('Correlation Matrix')
plt.show()
```
```

| | 0 |
|------------------------|---|
| GENDER | 0 |
| AGE | 0 |
| SMOKING | 0 |
| YELLOW_FINGERS | 0 |
| ANXIETY | 0 |
| PEER_PRESSURE | 0 |
| CHRONIC_DISEASE | 0 |
| FATIGUE | 0 |
| ALLERGY | 0 |





| | θ |
|-----------------------|----------|
| WHEEZING | 0 |
| ALCOHOL_CONSUMING | 0 |
| COUGHING | 0 |
| SHORTNESS_OF_BREATH | 0 |
| SWALLOWING_DIFFICULTY | 0 |
| CHEST_PAIN | 0 |
| LUNG_CANCER | 0 |

dtype: int64

GENDER

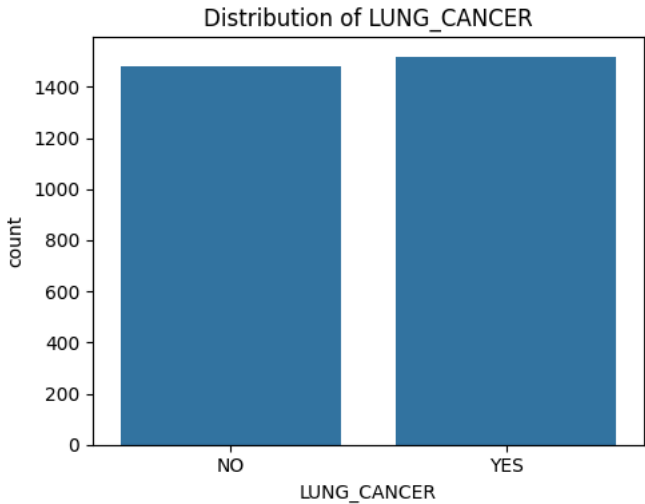
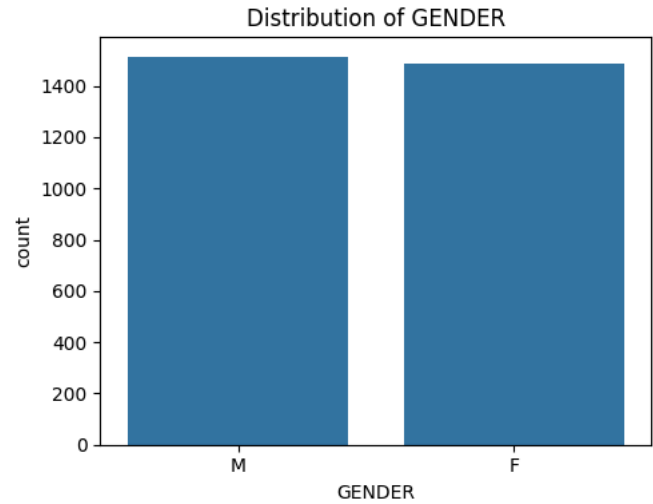
| | |
|---|------|
| M | 1514 |
| F | 1486 |

dtype: int64

LUNG_CANCER

| | |
|-----|------|
| YES | 1518 |
| NO | 1482 |

dtype: int64



| | AGE | SMOKING | YELLOW_FINGERS | ANXIETY | PEER_PRESSURE | CHRONIC_DISEASE | FATIGUE | ALLERGY | WHEEZING | ALCOHOL_CONSUMING | COUGHING | SHORTNESS_OF_BREATH | SWALLOWING_DIFF |
|-------|-------------|-------------|----------------|-------------|---------------|-----------------|-------------|-------------|-------------|-------------------|-------------|---------------------|-----------------|
| count | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.000000 | 3000.0 |
| mean | 55.169000 | 1.491000 | 1.514000 | 1.494000 | 1.499000 | 1.509667 | 1.489667 | 1.506667 | 1.497333 | 1.491333 | 1.510667 | 1.488000 | 1.4 |
| std | 14.723746 | 0.500002 | 0.499887 | 0.500047 | 0.500082 | 0.499990 | 0.499977 | 0.500039 | 0.500076 | 0.500008 | 0.499970 | 0.499939 | 0.4 |





| | AGE | SMOKING | YELLOW_FINGERS | ANXIETY | PEER_PRESSURE | CHRONIC_DISEASE | FATIGUE | ALLERGY | WHEEZING | ALCOHOL_CONSUMING | COUGHING | SHORTNESS_OF_BREATH | SWALLOWING_DIFF |
|-----|-----------|----------|----------------|----------|---------------|-----------------|----------|----------|----------|-------------------|----------|---------------------|-----------------|
| min | 30.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.0 |
| 25% | 42.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.0 |
| 50% | 55.000000 | 1.000000 | 2.000000 | 1.000000 | 1.000000 | 2.000000 | 1.000000 | 2.000000 | 1.000000 | 1.000000 | 2.000000 | 1.000000 | 1.0 |
| 75% | 68.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.0 |
| max | 80.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.000000 | 2.0 |



